

Standard model, published construction

Directed factor-mimicking portfolio catalogue

Abstract

This in-sample catalogue constructs one FMP for each configured field or composite and compares all factors on one factor- and control-complete monthly window. Factors are indexed by directed raw Sharpe, highest first. Direction is the sign that makes comparison-window Sharpe non-negative.

Methodology

Signals are observed at month-end, earn the next calendar month's simple return, and refresh monthly. Every factor must independently pass breadth, history, coverage, and non-degeneracy gates; none is silently dropped. One factor- and control-complete window fixes direction and Sharpe order and feeds all reported analyses. Risk Loadings include HAC t-statistics; Return Contributions are loading times annualized control mean, plus alpha and realized total. Rank diagnostics use 0/1/3/6/12/36-month horizons; available-history statistics are diagnostic only.

Construction Configuration

Setting	Effective value
Factors	4
Universe	shared security screen; refreshed monthly
Formation filters	price ≥ 1.0 ; 21d dollar volume $\geq$ none
Market-cap screen	none
Signal preparation	within-month winsorize, normalize, then demean
Composite	mean available normalized components; factor-specific coverage floor
Construction	3 effective profiles
Rebalance / return	month-end / next calendar month simple return
Costs	effective configured costs; default 0 bps
Minimum sample	20 stocks/month; 24 common months; 90% factor coverage
Comparison window	2008-11-01 to 2026-06-01; factor- and control-complete
Winsor settings	1%-99%
Controls	MKT, VAL, MOM, BAB, QMJ

Returns PCA

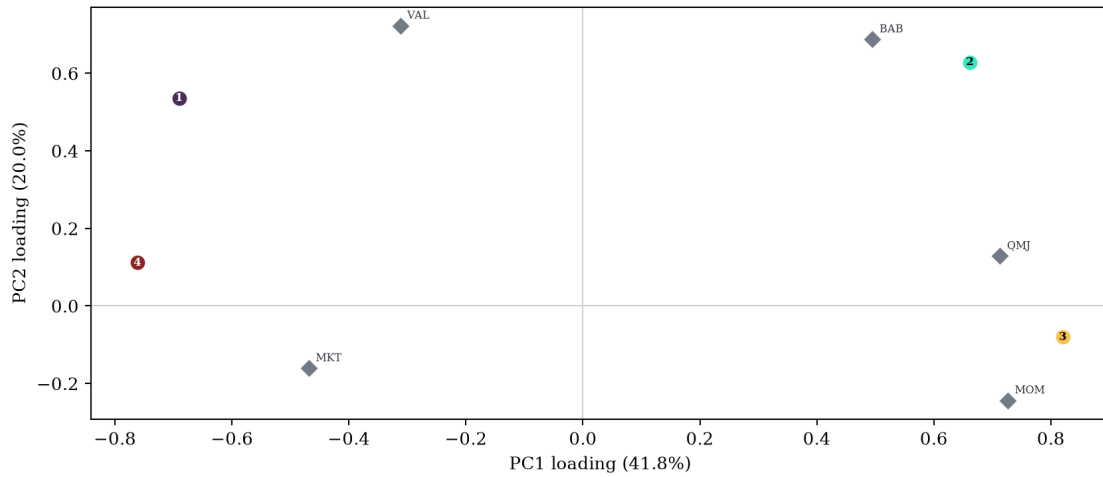


Figure: catalogue factors are numbered by directed-Sharpe index; attribution controls are named; $N=212$ common months.

Index

Idx	Factor	Dir.	N
1	VAL_rebuilt	+1	212
2	BAB_rebuilt	+1	212
3	MOM_rebuilt	+1	212
4	QMJ_rebuilt	-1	212

Returns

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
VAL_rebuilt	5.21%	9.82%	0.53	1.91	-39.10%	0.579	8.40%	9.82%
BAB_rebuilt	5.04%	9.87%	0.51	2.60	-17.02%	0.844	2.25%	9.87%
MOM_rebuilt	2.79%	14.05%	0.20	0.72	-55.10%	0.557	-2.43%	14.05%
QMJ_rebuilt	0.40%	11.39%	0.04	0.13	-46.16%	0.536	3.87%	11.39%

Risk Loadings

Factor	MKT	VAL	MOM	BAB	QMJ	Alpha	R2
VAL_rebuilt	-0.00 (-0.16)	0.41 (9.20)	-0.29 (-5.87)	-0.05 (-0.78)	-0.08 (-1.17)	8.15% (4.22)	0.579
BAB_rebuilt	-0.09 (-4.06)	0.01 (0.51)	0.07 (2.69)	0.72 (11.10)	0.06 (1.83)	2.50% (2.70)	0.844
MOM_rebuilt	-0.10 (-2.03)	0.04 (0.46)	0.61 (5.86)	0.06 (0.67)	0.28 (2.97)	-0.98% (-0.31)	0.565
QMJ_rebuilt	0.12 (2.26)	0.16 (3.26)	-0.10 (-2.27)	0.02 (0.30)	-0.50 (-6.47)	2.26% (0.95)	0.549

Return Contributions

Factor	MKT	VAL	MOM	BAB	QMJ	Alpha	Total
VAL_rebuilt	-0.07%	-0.55%	-1.67%	-0.22%	-0.43%	8.15%	5.21%
BAB_rebuilt	-1.32%	-0.02%	0.38%	3.14%	0.35%	2.50%	5.04%
MOM_rebuilt	-1.51%	-0.05%	3.48%	0.26%	1.60%	-0.98%	2.79%
QMJ_rebuilt	1.70%	-0.22%	-0.57%	0.08%	-2.84%	2.26%	0.40%

1. VAL_rebuilt

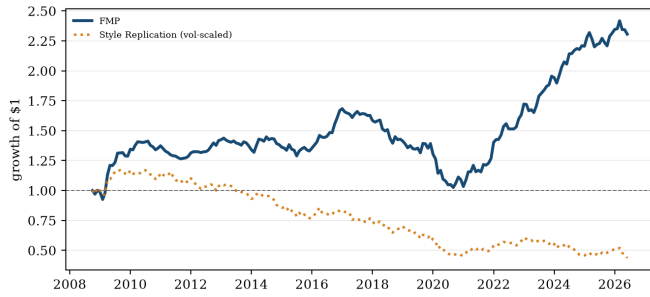


Figure: Common-window cumulative growth of \$1.

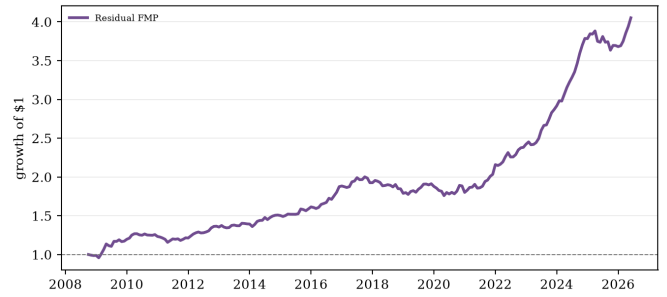


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
VAL_rebuilt	5.21%	9.82%	0.53	1.91	-39.10%	0.579	8.40%	9.82%

Factor	MKT	VAL	MOM	BAB	QMJ	Alpha	R2
VAL_rebuilt	-0.00 (-0.16)	0.41 (9.20)	-0.29 (-5.87)	-0.05 (-0.78)	-0.08 (-1.17)	8.15% (4.22)	0.579

Factor	MKT	VAL	MOM	BAB	QMJ	Alpha	Total
VAL_rebuilt	-0.07%	-0.55%	-1.67%	-0.22%	-0.43%	8.15%	5.21%

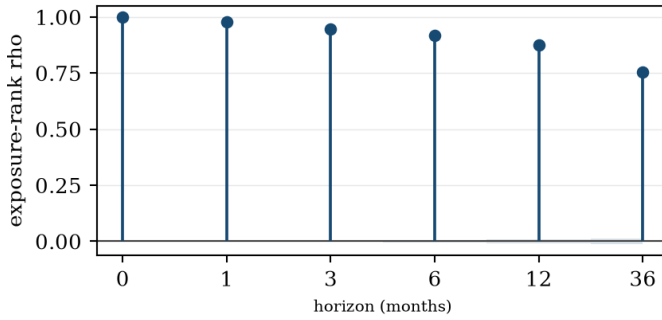


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

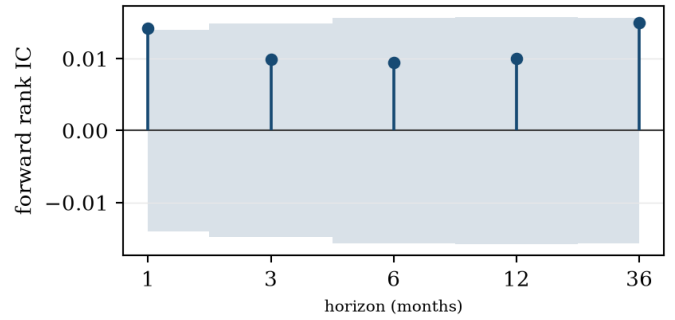


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	212
Signals	book_to_market
Component signs	book_to_market(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	255 / 212
Available window	2005-05-01 to 2026-07-01
Dropped: low breadth / common	51 / 43
Stocks, min / median / max	1741 / 2614 / 4495
Available-history Sharpe	0.39

2. BAB_rebuilt

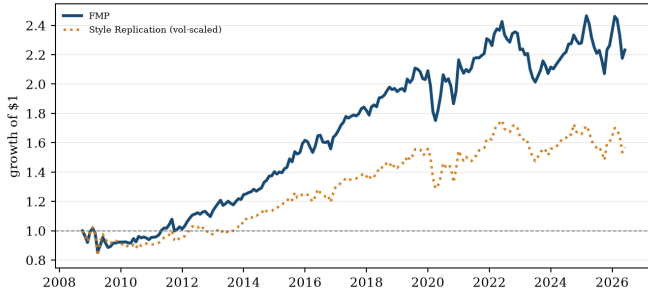


Figure: Common-window cumulative growth of \$1.

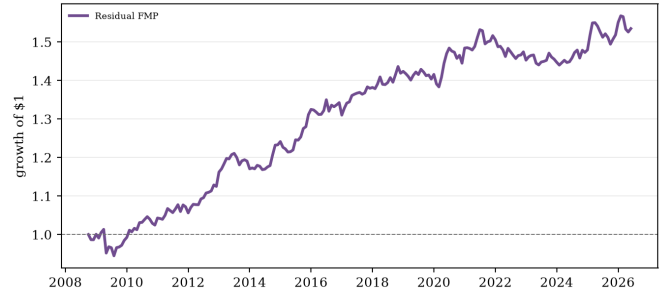


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
BAB_rebuilt	5.04%	9.87%	0.51	2.60	-17.02%	0.844	2.25%	9.87%

Factor	MKT	VAL	MOM	BAB	QMJ	Alpha	R2
BAB_rebuilt	-0.09 (-4.06)	0.01 (0.51)	0.07 (2.69)	0.72 (11.10)	0.06 (1.83)	2.50% (2.70)	0.844

Factor	MKT	VAL	MOM	BAB	QMJ	Alpha	Total
BAB_rebuilt	-1.32%	-0.02%	0.38%	3.14%	0.35%	2.50%	5.04%

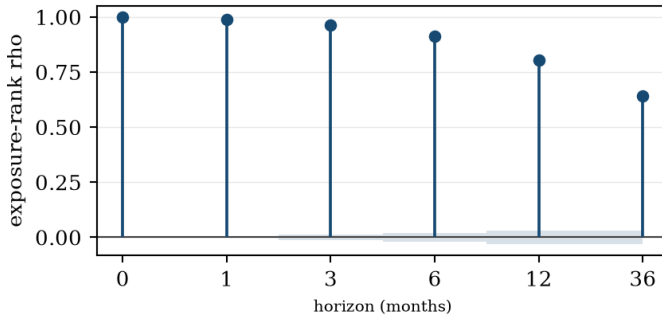


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

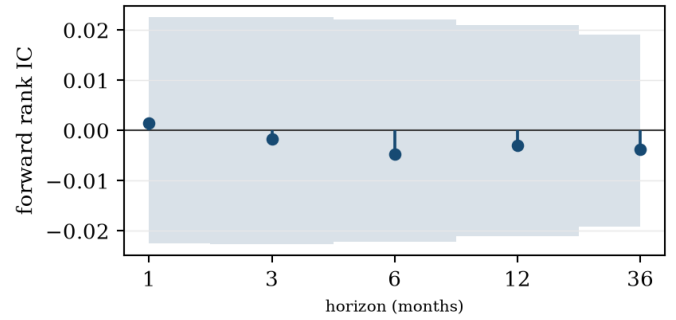


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	212
Signals	beta_shrunk_252v_1260c_VTI
Component signs	beta_shrunk_252v_1260c_VTI(-1)
Effective construction	rank/rank; 1%-99%; paper_bab; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	265 / 212
Available window	2004-07-01 to 2026-07-01
Dropped: low breadth / common	0 / 53
Stocks, min / median / max	1713 / 2486 / 4171
Available-history Sharpe	0.61

3. MOM_rebuilt

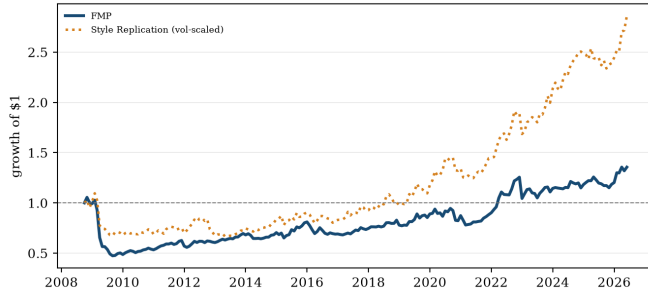


Figure: Common-window cumulative growth of \$1.

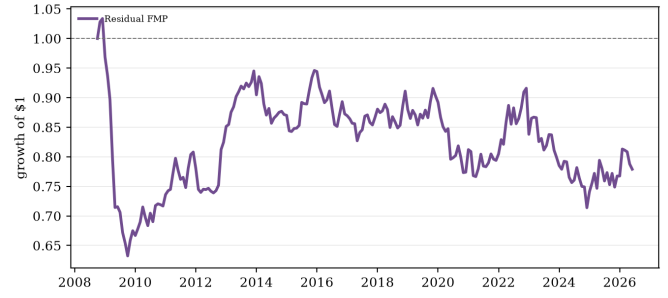


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
MOM_rebuilt	2.79%	14.05%	0.20	0.72	-55.10%	0.557	-2.43%	14.05%

Factor	MKT	VAL	MOM	BAB	QMJ	Alpha	R2
MOM_rebuilt	-0.10 (-2.03)	0.04 (0.46)	0.61 (5.86)	0.06 (0.67)	0.28 (2.97)	-0.98% (-0.31)	0.565

Factor	MKT	VAL	MOM	BAB	QMJ	Alpha	Total
MOM_rebuilt	-1.51%	-0.05%	3.48%	0.26%	1.60%	-0.98%	2.79%

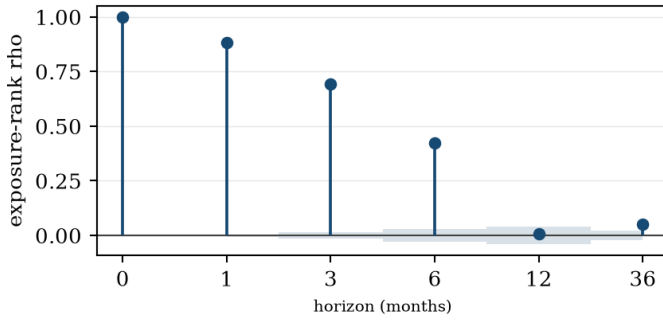


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

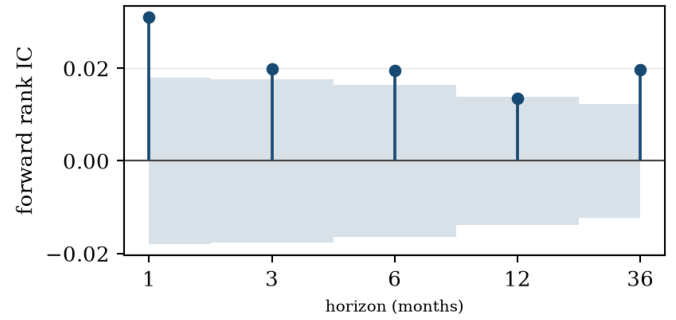


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	212
Signals	momentum_21_252
Component signs	momentum_21_252(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=False; group=none; cost=0bp
Components required	1 of 1
Available / comparison months	762 / 212
Available window	1963-02-01 to 2026-07-01
Dropped: low breadth / common	0 / 550
Stocks, min / median / max	1863 / 2705 / 4605
Available-history Sharpe	0.52

4. QMJ_rebuilt

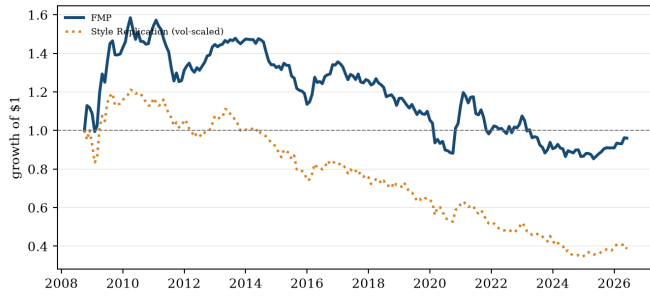


Figure: Common-window cumulative growth of \$1.

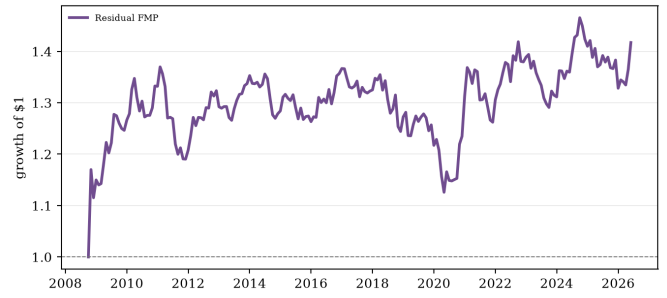


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
QMJ_rebuilt	0.40%	11.39%	0.04	0.13	-46.16%	0.536	3.87%	11.39%

Factor	MKT	VAL	MOM	BAB	QMJ	Alpha	R2
QMJ_rebuilt	0.12 (2.26)	0.16 (3.26)	-0.10 (-2.27)	0.02 (0.30)	-0.50 (-6.47)	2.26% (0.95)	0.549

Factor	MKT	VAL	MOM	BAB	QMJ	Alpha	Total
QMJ_rebuilt	1.70%	-0.22%	-0.57%	0.08%	-2.84%	2.26%	0.40%

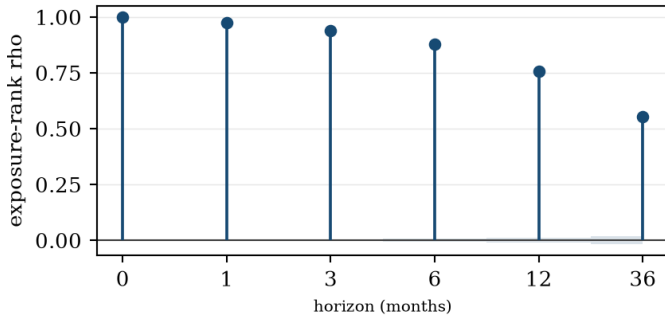


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

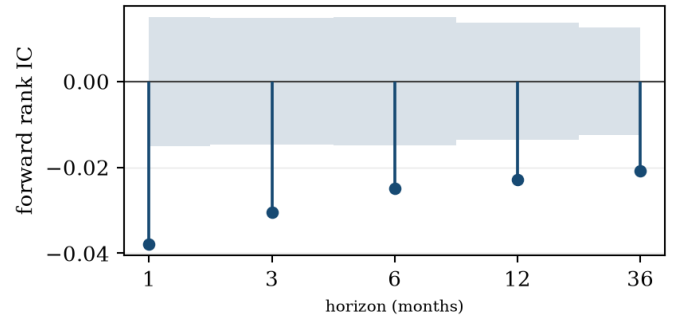


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	-1
Common months	212
Signals	gpoa, roe, roa, gross_margin, cfoa, accruals, growth_gpoa_5y, growth_roe_5y, growth_roa_5y, growth_cfoa_5y, growth_gmar_5y, growth_acc_5y, debt_to_assets, roe_volatility, altman_z, ivol_21d
Component signs	gpoa(+1), roe(+1), roa(+1), gross_margin(+1), cfoa(+1), accruals(-1), growth_gpoa_5y(+1), growth_roe_5y(+1), growth_roa_5y(+1), growth_cfoa_5y(+1), growth_gmar_5y(+1), growth_acc_5y(+1), debt_to_assets(-1), roe_volatility(-1), altman_z(+1), ivol_21d(-1)
Effective construction	rank/rank; 1%-99%; paper_qmj; LS; beta-neutral=False; group=none; cost=0bp
Components required	10 of 16
Available / comparison months	213 / 212
Available window	2008-11-01 to 2026-07-01
Dropped: low breadth / common	37 / 1
Stocks, min / median / max	34 / 2218 / 3609
Available-history Sharpe	0.01

Reproducibility

Run ID: 20260829T022753Z_standard-model-published-construction_f096c9c7cf.

Gold snapshot: version=20260826T131322Z.

Configuration hash: f096c9c7cf.

Symbols requested / loaded: all / 5857.

Seed: 0. Comparison months: 212, 2008-11-01 to 2026-06-01.

Command: C:/Users/User/Desktop/alpha/.venv/Scripts/python.exeC:/Users/User/Desktop/alpha/src/experiments/
factor_model_runner/run.py--configsrc/experiments/factor_model_runner/standard_rebuild_faithful.py--n0--seed0.

Direction is selected once from raw comparison-window Sharpe and then applied before every downstream regression, style calculation, plot, table, PCA coordinate, and index assignment. The run directory contains the source configuration, normalized configuration, manifest, canonical tables, monthly returns, PCA coordinates, plots, report source, and PDF.